Du Xujia

AI Application Engineer | **RAG** | **Agent Runtime** | **Backend Engineering**

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Target roles: AI / LLM Application Engineer (primary) | Python Backend / AI Backend Engineer (secondary)

Profile

AI application engineer working primarily in Python, with Java backend delivery experience across RAG retrieval, Agent orchestration, and service delivery. Since Sep 2025 I have independently taken on commercial and outsourced projects: the Jianzu Technology mini program, a paper retrieval task platform, and an IPA dialect transcription prototype for an end client at SUSTech. I also maintain open-source projects including a customer-service runtime and a Windows WeChat assistant, with delivery experience on an enterprise knowledge-base RAG/Agent workbench.

Client Delivery Experience

Jianzu Technology Shared-rental Mini Program

Commercial Project / Independent Contractor / Live Delivery | **2026.05 - Present**

Java 21 | Spring Boot | WeChat Pay v3 | MQTT | React | uni-app

- **Order state machine**: guarded payment, deposit-free, pickup, and return flows with explicit legal transitions, idempotency keys, and replay-safe callbacks so out-of-order or duplicated events cannot advance an order through an illegal path.
- **Platform onboarding and payment enablement**: owned domain registration and ICP filing; coordinated customer materials and applications for WeChat Pay and Alipay merchant accounts; completed WeChat Pay Score onboarding to open the payment capabilities required for launch.
- **Device reliability and safe unlock**: route devices to MQTT/HTTP/TCP by board model and protocol code; validate fresh online-and-closed snapshots before unlocking, then advance unresolved commands through retries, status queries, scheduled compensation, and manual exception handling.
- **Delivery**: coordinated domain, ICP, payment onboarding, production deployment, and delivery of the mini program, admin console, and channel console against 27 client requirements.

Shuge Technology (Shenzhen) | Paper Retrieval Task Platform

Outsourced Project / Independent Contractor / Client Delivery | **2026.01 - 2026.05**

Python | FastAPI | React | PostgreSQL | LISTEN/NOTIFY | SSE | Alembic

- **Task reuse and status sync**: built active-task reuse, editable titles, pending-worker diagnostics, and LISTEN/NOTIFY-driven SSE with snapshot/polling fallback while the client Worker only wrote to a shared tasks table.
- **Result, permission, and quota safety**: added guarded CSV/ZIP previews and controlled downloads; after successful completion, used transaction locking and unique quota events to constrain each task to at most one deduction.
- **Payment and delivery**: covered monthly/yearly orders, payment signature verification, pending-order reuse, QR refresh, and expiry closure; added health checks, frontend hosting, and one-command Windows start/deliver flows. Completed development independently and delivered the code to the client (the second project delivered to this client).

Shuge Technology (Shenzhen) | IPA Dialect Speech Transcription Prototype

Outsourced Project / Technical Consultant / Client Delivery | **End client: SUSTech** | **2025.11 - 2025.12**

Flask | PyTorch | Transformers | Torchaudio | FFmpeg | SSE

- **Scope clarification**: led requirement clarification and prototype decomposition under ambiguous requirements, narrowing scope to an acceptance-ready MVP (local transcription + batch review + Excel export).
- **Local ASR and review flow**: delivered a locally runnable dialect-to-IPA prototype with FFmpeg/Torchaudio normalization; connected Excel rules for initials, finals, and tones with template validation and export. The client accepted the prototype on-site using real audio samples; logs trace 2,380 recognitions, 60 batch jobs, and a largest batch of 500 files.

Independent / Open Source Projects

AI Customer Service Runtime

Independent Developer / Open Source | **2026.02 - Present**

Python 3.13 | FastAPI | Pydantic v2 | OpenAI Responses API adapter | custom controlled Agent harness / Tool Catalog | RAG | Qdrant / Pinecone / Milvus | SQLite / Redis | RTC WebSocket | Docker

- **Controlled Agent tool flow**: built a customer-service runtime for knowledge Q&A, business lookups, and human handoff; separated channel adapters, host bridge, core engine, business tools, and model providers. The Tool Catalog scopes tools by tenant, industry, and channel and validates required parameters.
- **Recoverable execution and human gates**: implemented explicit steps, allowed_tools, max_steps, fail-stop behavior, checkpoint/resume, approval queues, and step traces; handoff supports local / SQLite / optional Redis sorted-set + WATCH/MULTI atomic claims. The verification path covers allowlists, step limits, fail-stop behavior, approval blocking, and checkpoint recovery (a controlled sequential workflow, not open-ended autonomous planning).
- **Knowledge governance and verification**: used KnowledgeService for tenants, documents, chunks, and versions; supported vector-store adapters, snapshot activation, health checks, citations, and local RAG evaluation with regression gates.

WeChat Intelligent Assistant

Independent Developer / Open Source | **2025.12 - Present**

WCFerry | Quart | Electron | SQLite | OpenAI-compatible API | RAG

- **Desktop transport and direct runtime**: abstracted BaseTransport for Windows WeChat version, permission, and message-channel differences; migrated the synchronous reply path to direct OpenAI-compatible /chat/completions and /embeddings, separating first response from background fact extraction, vector writes, profile updates, and corpus sync.
- **Memory, productization, and verification**: built SQLite short-term memory and vector storage for runtime/exported-corpus RAG, with optional hybrid search and lightweight fallback; added model auth, hot reload, readiness endpoints, sanitized diagnostics, backup/restore, offline evaluation, and Windows release gates.

Internship Experience

Chinasoft International

2025.04 - 2025.09

Enterprise Knowledge-base RAG / Agent Workbench (RAG-QA System) | Backend / Full-stack Engineer

FastAPI | LangGraph | PostgreSQL FTS | Qdrant | FastEmbed | Vue 3 | MinIO

- **Retrieval and evidence**: combined PostgreSQL full-text search, vector retrieval, and structural signals with query rewrite, weighted RRF, and reranking; returned citations, grounding scores, latency breakdowns, and trace IDs, with offline retrieval comparisons guarding against regression.
- **Document-level access filtering**: stored only document allowed_departments in the knowledge base, never per-person permissions; derived the caller's department from the signed JWT CurrentUser.department and filtered unauthorized documents before all retrieval paths, including explicit document_ids.
- **Recoverable Q&A and governance**: moved orchestration to LangGraph with checkpoint, interrupt/resume, human clarification, and step_events; built ingest, chunk split/merge/disable, version clarification, and retrieve/debug workbenches.
- **Evaluation and release gates**: added model-health checks, context/cost budgets, PII detection, TTFT tracking, semantic caching, and regression gates. Built an offline evaluation evidence pack with a version-checked baseline and regression gates, and wired a pytest group runner into CI so evaluation assets stay traceable and reproducible.

National Clinical Research Center for Orthopedics

2024.08 - 2024.10

SubMed Medical Literature Search Mini Program | Backend Developer Intern

- **Medical search and analysis loop**: participated in API design and integration for AI search, voice input, paper details, recommendations, deep analysis, and sharing across the SubMed mini program.
- **Continuous research feed**: designed subscription settings and push history so users could track papers by keyword, impact factor, partition, and frequency.

China Unicom, Shaanxi Branch

2024.05 - 2024.08

Touchpoint Enablement Center | Shaanxi Operations Platform | Data Platform | Backend Developer Intern

- **Query and analytics governance**: participated in OLTP/OLAP separation, ClickHouse aggregation, and SQL/index/cache tuning for activity statistics and operations data.
- **Migration and release governance**: participated in migration checks, xxl-job daily updates, and CI/CD release/rollback paths with business continuity and traceability in mind.

Additional Projects

EasyCloudPan

Full-stack Developer / Open Source | **2024.04 - 2026.02**

Spring Boot | React | PostgreSQL | Redis | MinIO

- **Cloud storage**: built chunked and resumable uploads, object-storage consistency checks, signed access, tenant isolation, caching, and health monitoring.

Jointown Four-way Shuttle Path Planning System

Algorithm R&D / Prototype | **2023.05 - 2023.07**

Python | A* / CBS-inspired conflict handling | PyQt5 | multiprocessing

- **Warehouse path planning**: modeled a grid, implemented A* with priority-based waiting for multi-vehicle conflicts, and built a PyQt5 step-debug simulator.

AI Application Skills

- **RAG and Agent applications**: parsing and chunking, embeddings, hybrid retrieval, reranking, citation grounding, evidence-based refusal, query rewrite, tool calling, allowlists, checkpointing, interrupt / resume, and human handoff
- **Evaluation and service delivery**: offline retrieval and citation evaluation, regression gates, FastAPI, SSE, model routing, fallback, semantic caching, token tracking, billing reconciliation, and PII redaction

Technical Skills

- **Backend and engineering**: Python, Java, FastAPI, Flask, Quart, Spring Boot, MyBatis-Flex, Spring Security, REST APIs, SSE, WebSocket, MQTT, authentication, retries, idempotency, transactions, distributed locks, state machines, MySQL, PostgreSQL, Redis, Linux, Docker, Git, CI/CD, and Prometheus / Grafana

Education Nanyang Institute of Technology · Data Science & Big Data · B.S. · 2021.09 - 2025.06 · GPA 3.8/4.5, Top 5%